

Mathematical Foundations of Political Methodology

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Lecture 6: Transforms

- Proof that if $X_1 \dots X_N$ are MVN and $\text{cov}(X_i, X_j) = 0$ then independent.
- Change of Coordinates
- Moment Generating Functions

Definition

$\mathbf{X} = (X_1, X_2, \dots, X_N)$ is a vector of random variables. If $\mathbf{X}$ has pdf

$$f(\mathbf{x}) = (2\pi)^{-N/2} \det(\mathbf{\Sigma})^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \mathbf{\Sigma}(\mathbf{x} - \boldsymbol{\mu})\right)$$

$\mathbf{X}$ is a *Multivariate Normal* Distribution,

$$\mathbf{X} \sim \text{Multivariate Normal}(\boldsymbol{\mu}, \mathbf{\Sigma})$$

Proposition

Suppose $\mathbf{X} \sim \text{Multivariate Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. where $\mathbf{X} = (X_1, X_2, \dots, X_N)$. If $\text{cov}(X_i, X_j) = 0$, then X_i and X_j are independent

Proof.

Suppose $\mathbf{X} = (X_1, \dots, X_N)$ are MVN distributed

Partition these into two groups $\mathbf{X}_A$ and $\mathbf{X}_B$, with means μ_A and μ_B , and variance covariance

$$\Sigma = \begin{pmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_N) \\ \text{cov}(X_2, X_1) & \text{var}(X_2) & \dots & \text{cov}(X_2, X_N) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_N, X_1) & \text{cov}(X_N, X_2) & \dots & \text{var}(X_N) \end{pmatrix} = \begin{pmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{pmatrix}$$

Now assume: $\Sigma_{AB} = \Sigma_{BA} = 0$



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$$\begin{bmatrix} \mathbf{x}_A^T - \mu_A^T & \mathbf{x}_B^T - \mu_B^T \end{bmatrix} \begin{bmatrix} \Sigma_{AA} & 0 \\ 0 & \Sigma_{BB} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{x}_A - \mu_A \\ \mathbf{x}_B - \mu_B \end{bmatrix}$$



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So $\exp\{(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)\}$ factorizes into :

$$\exp\{(\mathbf{x}_A - \mu_A)^T \Sigma_{AA}^{-1} (\mathbf{x}_A - \mu_A)\} * \exp\{(\mathbf{x}_B - \mu_B)^T \Sigma_{BB}^{-1} (\mathbf{x}_B - \mu_B)\}$$

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so :

$$f(\mathbf{x}) = g(\mathbf{x}_A)h(\mathbf{x}_B)$$

Change of Coordinates

Proposition

Suppose X is a random variable and $Y = g(X)$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ that is a monotonic function.

Define $g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ such that $g^{-1}(g(X)) = X$ and is differentiable. Then,

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{\partial g^{-1}(y)}{\partial y} \right| \text{ if } y = g(x) \text{ for some } x \\ &= 0 \text{ otherwise} \end{aligned}$$

Change of Coordinates

Proof.

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$$\begin{aligned}F_Y(y) &= P(Y \leq y) \\&= P(g(X) \leq y) \\&= P(X \leq g^{-1}(y)) \\&= F_X(g^{-1}(y))\end{aligned}$$

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$$\begin{aligned}\frac{\partial F_Y(y)}{\partial y} &= \frac{\partial F_X(g^{-1}(y))}{\partial y} \\&= f_X(g^{-1}(y)) \frac{\partial g^{-1}(y)}{\partial y}\end{aligned}$$

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Then this is a pdf because $\frac{\partial g^{-1}(y)}{\partial y} > 0$.

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$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{\partial g^{-1}(Y)}{\partial y} \right|$$

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We've used this to derive many of the pdfs

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- Normal distribution
- Chi-Squared Distribution

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Suppose $Z \sim \text{Normal}(0, 1) \leftarrow f_x(x)$.

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Then if $X = \sum_{i=1}^N Z^2$

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Approximating functions and second order conditions

Theorem

Taylor's Theorem Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x)$ is infinitely differentiable function. Then, the Taylor expansion of $f(x)$ around a is given by

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$$

Example Function

Suppose $a = 0$ and $f(x) = e^x$. Then,

$$f'(x) = e^x$$

$$f''(x) = e^x$$

$$\vdots \quad \vdots \quad \vdots$$

$$f^n(x) = e^x$$

This implies

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \dots + \frac{x^n}{n!} + \dots$$

Moment Generating Functions

Definition

Suppose X is a random variable with pdf f . Define,

$$E[X^n] = \int_{-\infty}^{\infty} x^n f(x) dx$$

We will call X^n the n^{th} moment of X

- By this definition $\text{var}(X) = \text{Second Moment} - \text{First Moment}^2$
- We are assuming that the integral converges

Moment Generating Functions

Proposition

Suppose X is a random variable with pdf $f(x)$. Call $M(t) = E[e^{tX}]$,

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \end{aligned}$$

We will call $M(t)$ the moment generating function, because:

$$\left. \frac{\partial^n M(t)}{\partial^n t} \right|_0 = E[X^n]$$

(Assuming that we can interchange derivative and integral)

Moment Generating Functions

Proof.

Recall the Taylor Expansion of e^{tX} at 0,



Moment Generating Functions

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Recall the Taylor Expansion of e^{tX} at 0,

$$e^{tX} = 1 + tX + \frac{t^2 X^2}{2!} + \frac{t^3 X^3}{3!} + \dots$$



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Then,

$$E[e^{tX}] = 1 + tE[X] + \frac{t^2}{2!}E[X^2] + \frac{t^3}{3!}E[X^3] + \dots$$



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Recall the Taylor Expansion of e^{tX} at 0,

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Differentiate once:



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Differentiate once:

$$\begin{aligned}\frac{\partial M(t)}{\partial t} &= 0 + E[X] + \frac{2t}{2!}E[X^2] + \dots \\ M'(0) &= 0 + E[X] + 0 + 0 \dots\end{aligned}$$



Proof.

Differentiate n times



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Differentiate n times

$$\frac{\partial^n M(t)}{\partial^n t} = 0 + 0 + 0 + \dots + \frac{n \times n - 1 \times \dots 2 \times t^0 E[X^n]}{n!} + \frac{n! t E[X^{n+1}]}{(n+1)!} + \dots$$



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Differentiate n times

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Evaluated at 0, yields $M^n(0) = E[X^n]$



Proof.

Differentiate n times

$$\begin{aligned}\frac{\partial^n M(t)}{\partial^n t} &= 0 + 0 + 0 + \dots + \frac{n \times n - 1 \times \dots \times 2 \times t^0 E[X^n]}{n!} + \frac{n! t E[X^{n+1}]}{(n+1)!} + \dots \\ &= \frac{n! E[X^n]}{n!} + \frac{n! t E[X^{n+1}]}{(n+1)!} + \dots\end{aligned}$$

Evaluated at 0, yields $M^n(0) = E[X^n]$



- If two random variables, X and Y have the same moment generating functions, then $F_X(x) = F_Y(y)$ for **almost all** x .

The Moments of the Normal Distribution

Suppose $Z \sim N(0, 1)$.

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Proposition

Suppose X_i are a sequence of independent random variables. Define

$$Y = \sum_{i=1}^N X_i$$

Then

$$M_Y(t) = \prod_{i=1}^N M_{X_i}(t)$$

Proof.

$$M_Y(t) = E[e^{tY}]$$



Proof.

$$\begin{aligned} M_Y(t) &= E[e^{tY}] \\ &= E[e^{t \sum_{i=1}^N X_i}] \end{aligned}$$



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Inequalities and Limit Theorems

Limit Theorems

- What happens when we consider a long sequence of random variables?
- What can we reasonably infer from data?
 - Laws of large numbers: averages of random variables converge on expected value?
 - Central Limit Theorems: sum of random variables have normal distribution?
- We'll focus on intuition for both, but we'll prove some stuff too.

Review Session

Weak Law of Large Numbers

Proof plan:

- Markov's Inequality
- Chebyshev's Inequality
- Weak Law of Large Numbers

Markov's Inequality

Proposition

Suppose X is a random variable that takes on non-negative values. Then, for all $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Markov's Inequality

Proof.

For $a > 0$,



Markov's Inequality

Proof.

For $a > 0$,

$$E[X] = \int_0^{\infty} xf(x)dx$$



Markov's Inequality

Proof.

For $a > 0$,

$$\begin{aligned} E[X] &= \int_0^{\infty} xf(x)dx \\ &= \int_0^a xf(x)dx + \int_a^{\infty} xf(x)dx \end{aligned}$$



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$$E[X] \geq \int_a^{\infty} xf(x)dx \geq \int_a^{\infty} af(x)dx = aP(X \geq a)$$



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Because $X \geq 0$,

$$\begin{aligned} E[X] &\geq \int_a^{\infty} xf(x)dx \geq \int_a^{\infty} af(x)dx = aP(X \geq a) \\ \frac{E[X]}{a} &\geq P(X \geq a) \end{aligned}$$



Chebyshev's Inequality

Proposition

If X is a random variable with mean μ and variance σ^2 , then, for any value $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Chebyshev's Inequality

Proof.

Define the random variable

$$Y = (X - \mu)^2$$

Where $\mu = E[X]$.



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Where $\mu = E[X]$.

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Applying the inequality:

$$\begin{aligned} P(Y \geq k^2) &\leq \frac{E[Y]}{k^2} \\ P((X - \mu)^2 \geq k^2) &\leq \frac{E[(X - \mu)^2]}{k^2} \end{aligned}$$



Chebyshev's Inequality

Proof.

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$$Y = (X - \mu)^2$$

Where $\mu = E[X]$.

Then we know Y is a non-negative random variable. Set $a = k^2$.

Applying the inequality:

$$P(Y \geq k^2) \leq \frac{E[Y]}{k^2}$$

$$P((X - \mu)^2 \geq k^2) \leq \frac{E[(X - \mu)^2]}{k^2}$$

$$P((X - \mu)^2 \geq k^2) \leq \frac{\sigma^2}{k^2}$$



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Sequence of Random Variables

Sequence of Independent and Identically, Distributed Random variables.

- Sequence: $X_1, X_2, \dots, X_n, \dots$
- Think of a sequence as sampled **data**:
 - Suppose we are drawing a sample of N observations
 - Each observation will be a random variable, say X_i
 - With realization x_i

Mean/Variance of Sample Mean

Proposition

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and variance σ^2 . Let $\bar{X}_n$ be the sample mean. Then $E[\bar{X}_n] = \mu$ and $\text{var}(\bar{X}_n) = \frac{\sigma^2}{n}$

Proof.



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Proof.

$$E[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^n E[X_i]$$



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Proof.

$$\begin{aligned} E[\bar{X}_n] &= \frac{1}{n} \sum_{i=1}^n E[X_i] \\ &= \frac{1}{n} n\mu = \mu \end{aligned}$$



Mean/Variance of Sample Mean

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$$\text{var}(\bar{X}_n) = \frac{1}{n^2} \text{var}\left(\sum_{i=1}^n X_i\right)$$

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Weak Law of Large Numbers

Proposition

Suppose $X_1, X_2, \dots, X_n$ is a random sample from a distribution with mean μ and $\text{Var}(X_i) = \sigma^2$. Then, for all $\epsilon > 0$,

$$P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \mu \right| \geq \epsilon \right\} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Weak Law of Large Numbers

Proof.

From our previous proposition



Weak Law of Large Numbers

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From our previous proposition

$$\frac{E[X_1 + X_2 + \cdots + X_n]}{n} = \frac{\sum_{i=1}^n E[X_i]}{n} = \mu$$



Weak Law of Large Numbers

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From our previous proposition

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Further,



Weak Law of Large Numbers

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From our previous proposition

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Further,

$$E\left[\left(\frac{\sum_{i=1}^n X_i - n\mu}{n}\right)^2\right] = \frac{\text{Var}(X_1 + X_2 + \cdots + X_n)}{n^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2} = \frac{\sigma^2}{n}$$



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Apply Chebyshev's Inequality:



Weak Law of Large Numbers

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From our previous proposition

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Apply Chebyshev's Inequality:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| \geq \epsilon\right\} \leq \frac{\sigma^2}{n\epsilon^2}$$



Suppose $X_1, X_2, \dots$ are iid normal distributions,

$$X_i \sim \text{Normal}(0, 10)$$

$$P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \mu \right| \geq 0.1 \right\} \text{ as } n \rightarrow \infty$$

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Suppose we want to guarantee that we have at most a 0.01 probability of being more than 0.1 away from the true μ . How big do we need n ?

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$$\begin{aligned} 0.01 &= \frac{10}{n(0.1^2)} \\ n &= \frac{1000}{0.01} \end{aligned}$$

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$$X_i \sim \text{Normal}(0, 10)$$

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Suppose we want to guarantee that we have at most a 0.01 probability of being more than 0.1 away from the true μ . How big do we need n ?

$$\begin{aligned} 0.01 &= \frac{10}{n(0.1^2)} \\ n &= \frac{1000}{0.01} \\ n &= 100,000 \end{aligned}$$